



US012402657B2

(12) **United States Patent**
Fraser et al.

(10) **Patent No.:** **US 12,402,657 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) **ELECTRICAL VAPOR PROVISION SYSTEM WITHOUT WIRED ELECTRICAL CONNECTION BETWEEN A CONTROL UNIT AND A CARTRIDGE**

(58) **Field of Classification Search**
CPC A24F 40/42; A24F 40/46; A24F 40/465
See application file for complete search history.

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 21 days.

(21) Appl. No.: **18/539,716**

(22) Filed: **Dec. 14, 2023**

Notice of Allowance received for Chinese Patent Application No. 202110630668.4, mailed on Jun. 10, 2024, 6 pages (2 pages of English Translation and 4 pages of Official Copy).

(Continued)

(65) **Prior Publication Data**

US 2024/0108062 A1 Apr. 4, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/456,446, filed on Nov. 24, 2021, now Pat. No. 11,882,877, which is a
(Continued)

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(30) **Foreign Application Priority Data**

Jun. 29, 2015 (GB) 1511359

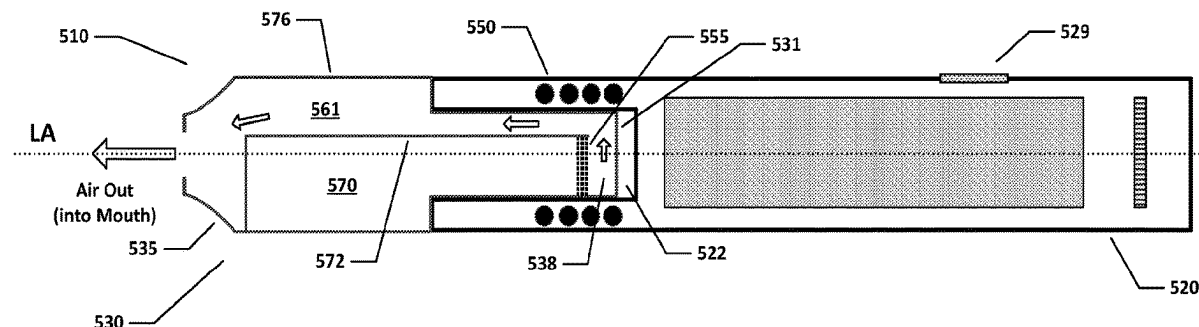
(51) **Int. Cl.**
A24F 40/42 (2020.01)
A24F 40/465 (2020.01)
(Continued)

(57) **ABSTRACT**

An electronic vapor provision system includes a control unit and a cartridge configured to engage with and disengage from one another substantially along a longitudinal axis. The cartridge includes a reservoir of liquid to be vaporized, and the control unit includes a battery and a tube portion disposed along said longitudinal axis. The system further includes an induction heating assembly comprising a drive coil and a heater element. The heater element is located in the cartridge for vaporizing the liquid. The cartridge is configured to feed liquid from the reservoir onto the heater element for vaporization. The drive coil is located in the tube portion of the control unit and disposed about the longitudinal axis. The cartridge is located at least partly within the tube portion when engaged with the control unit, such that the heater element is within the drive coil.

(52) **U.S. Cl.**
CPC **A24F 40/42** (2020.01); **A24F 40/465** (2020.01); **A24F 40/10** (2020.01); **A24F 40/30** (2020.01);
(Continued)

10 Claims, 6 Drawing Sheets



Related U.S. Application Data

continuation of application No. 15/739,037, filed as application No. PCT/GB2016/051766 on Jun. 15, 2016, now Pat. No. 11,185,110.

(51) Int. Cl.

A24F 40/10 (2020.01)

A24F 40/30 (2020.01)

A24F 40/44 (2020.01)

A61M 11/04 (2006.01)

A61M 15/06 (2006.01)

(52) U.S. Cl.

CPC *A24F 40/44* (2020.01); *A61M 11/042* (2014.02); *A61M 15/06* (2013.01)

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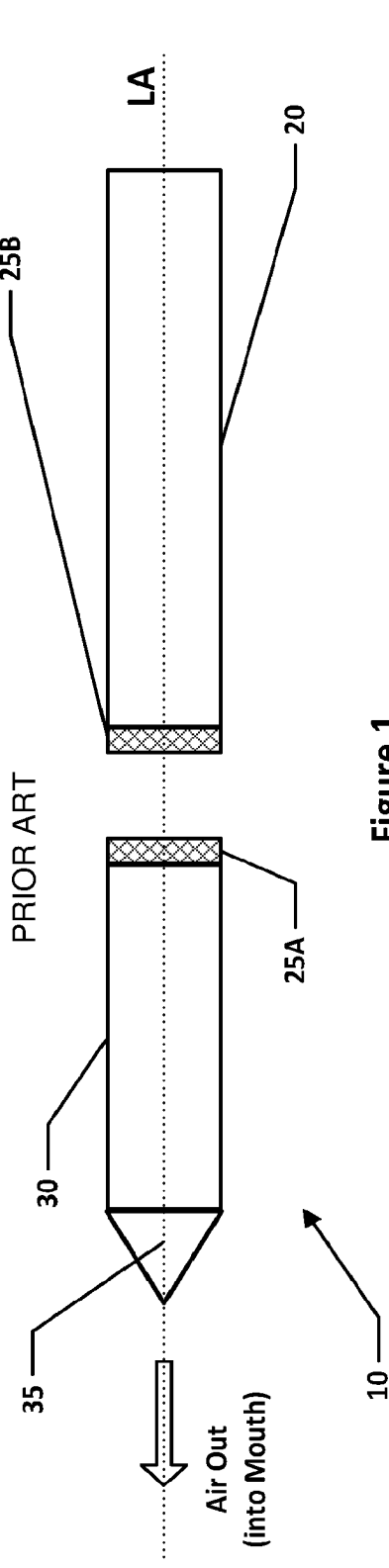


Figure 1

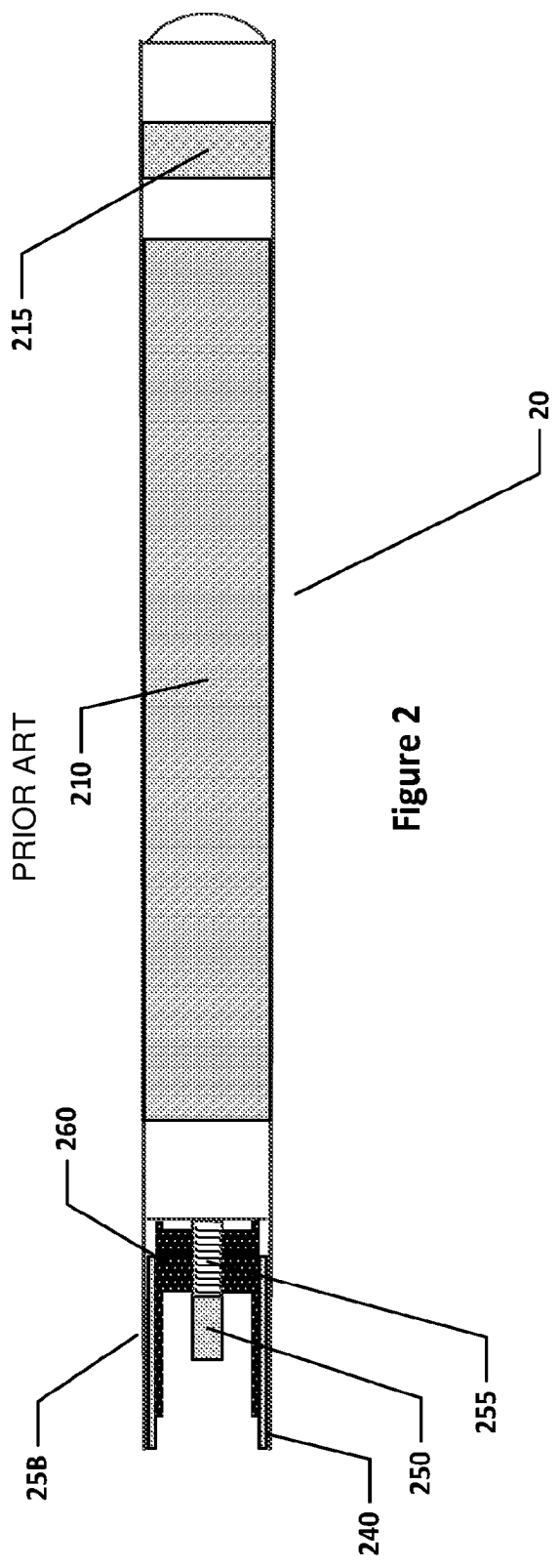


Figure 2

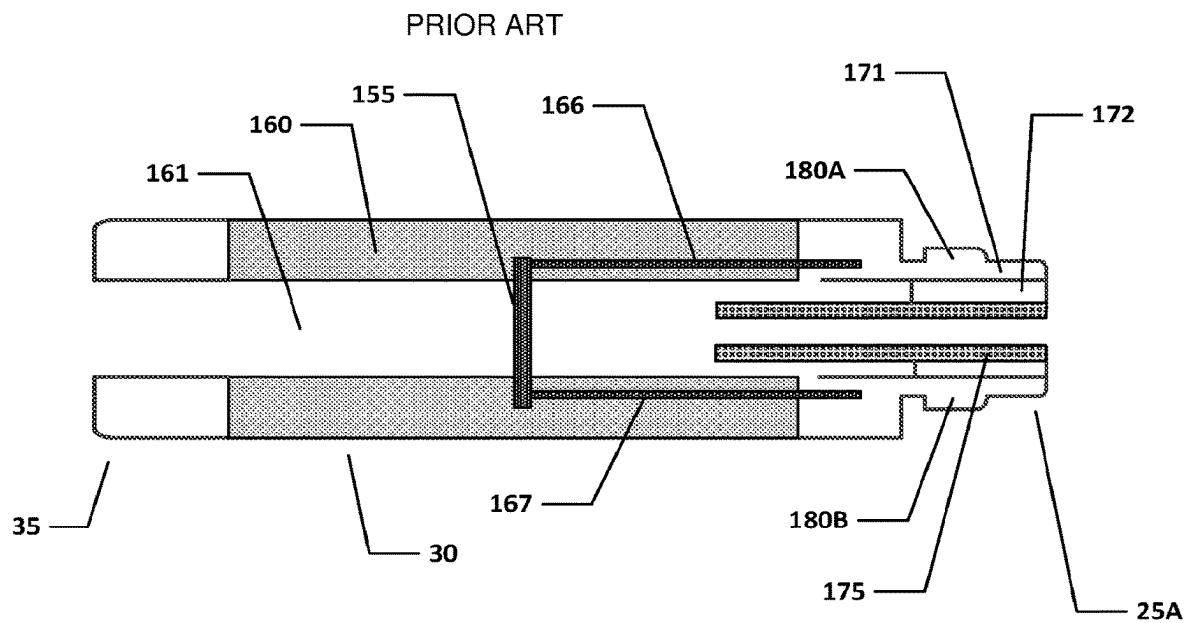


Figure 3

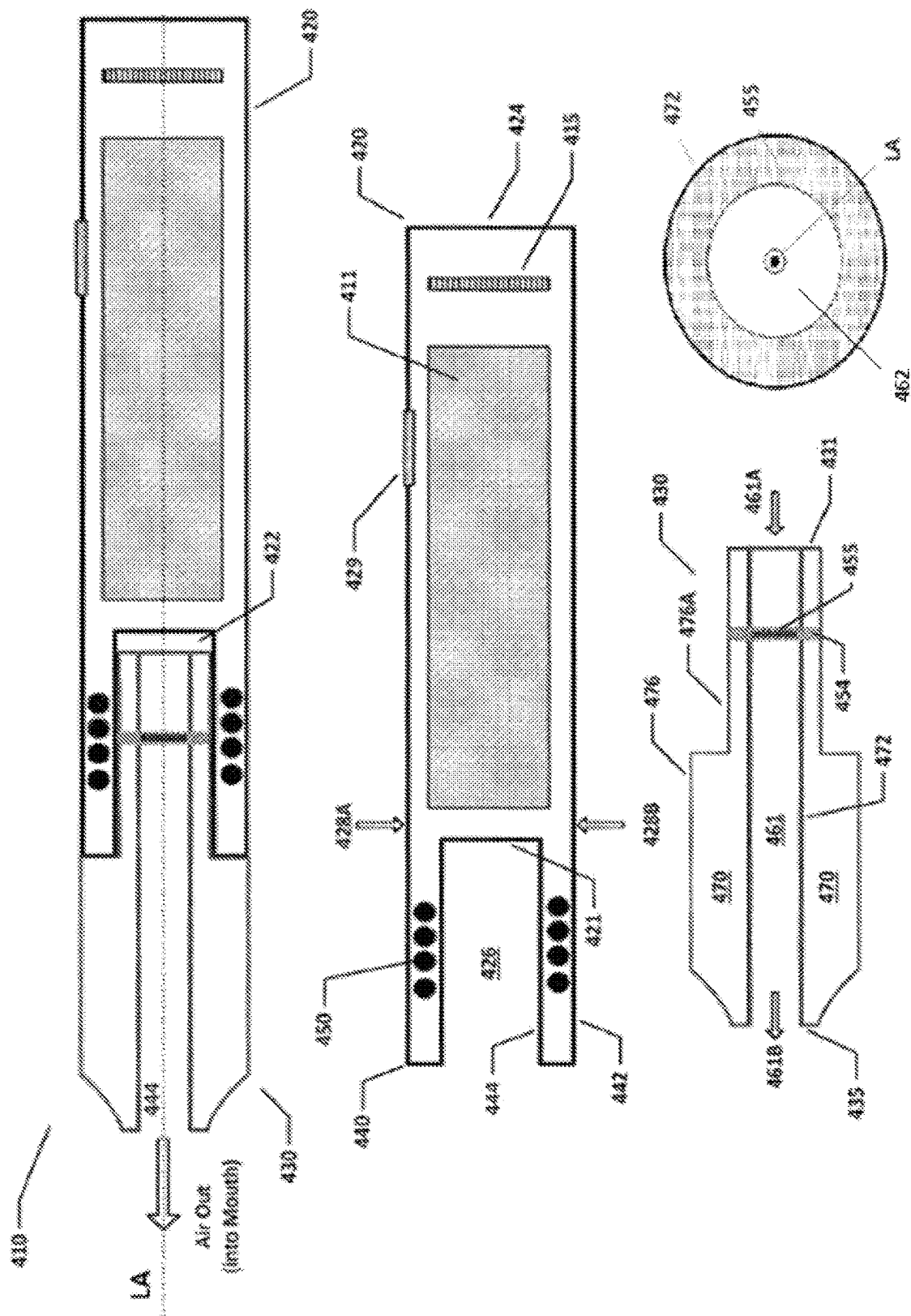


Figure 4

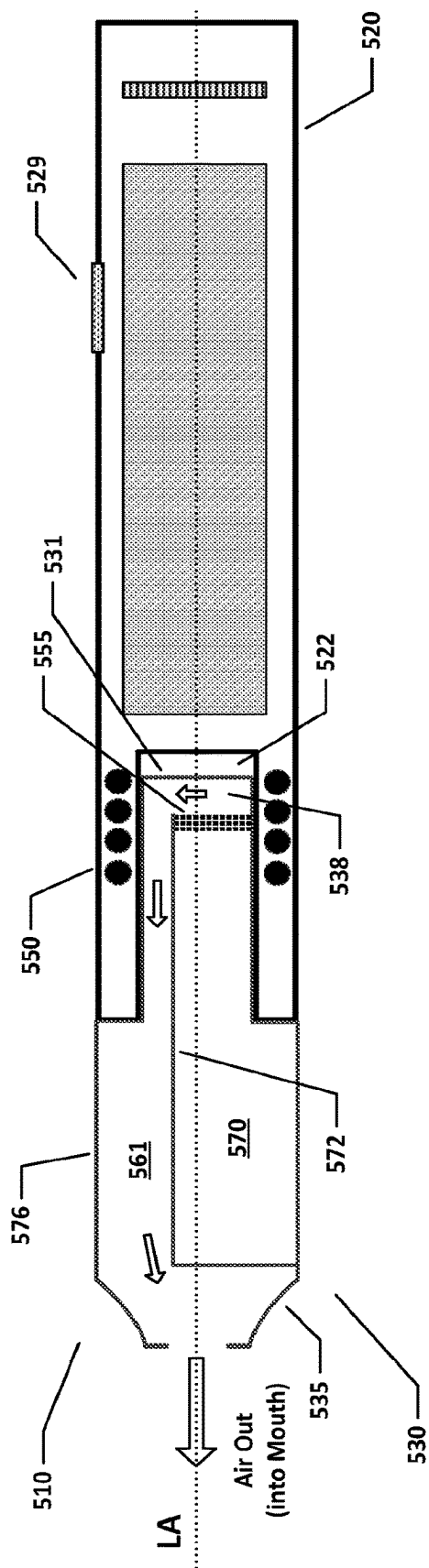


Figure 5

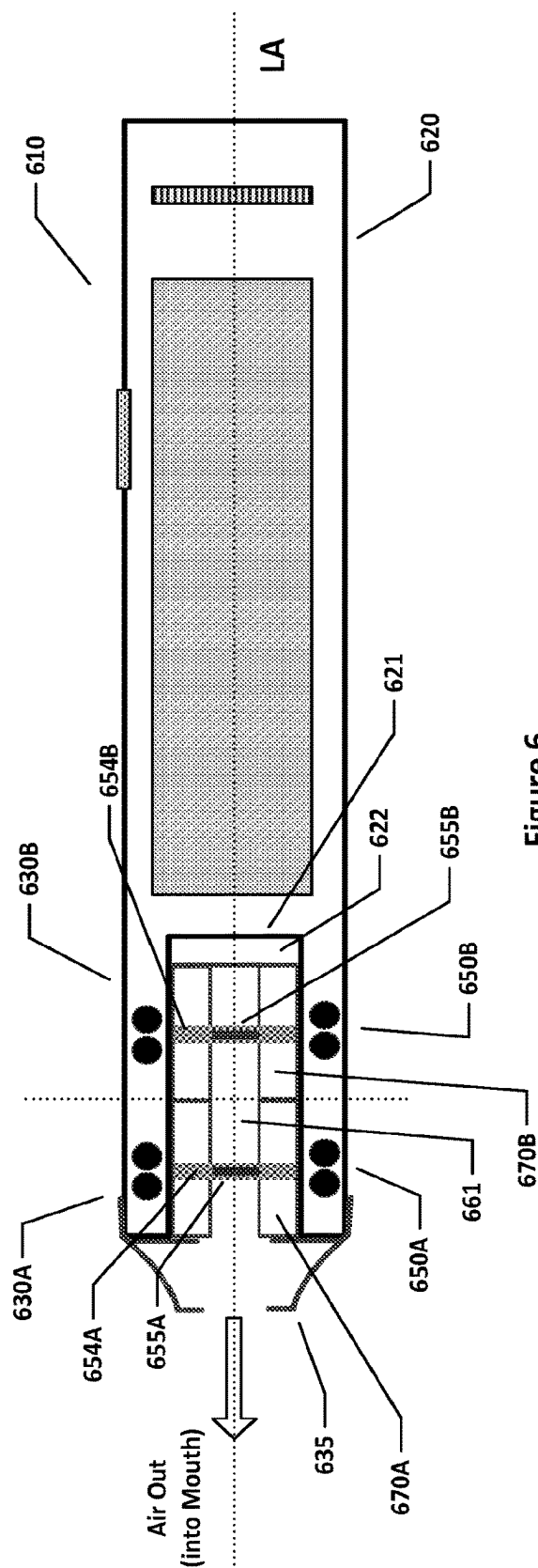


Figure 6

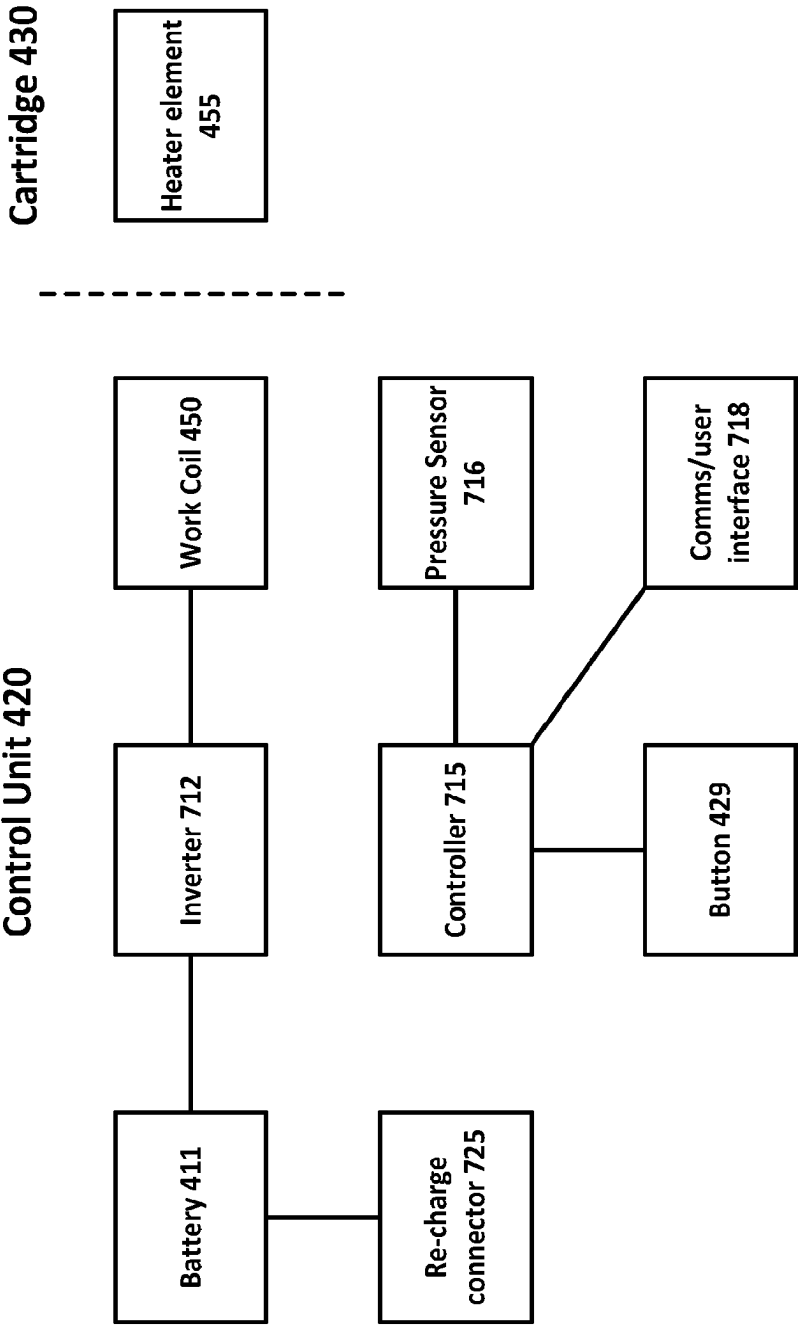


Figure 7

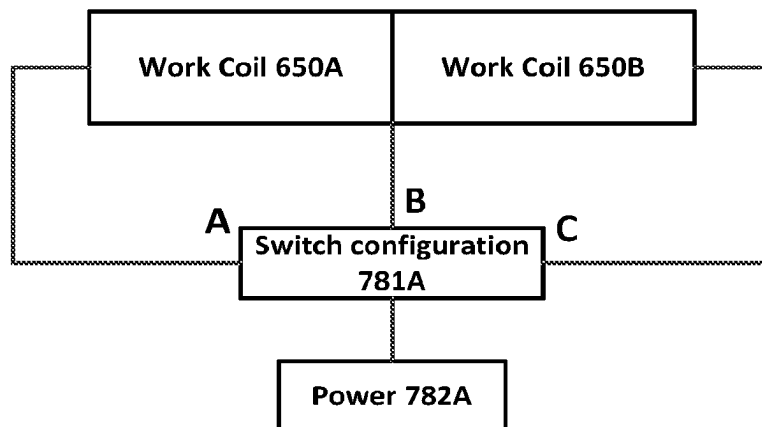


Figure 7A

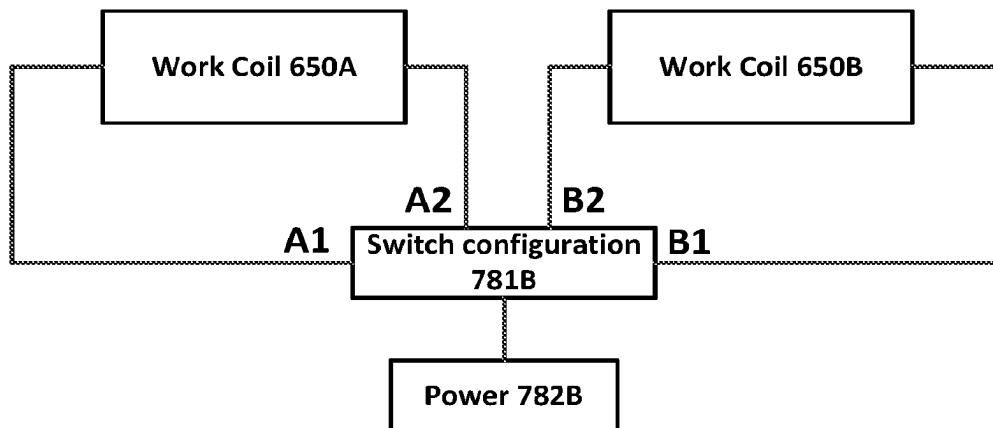


Figure 7B

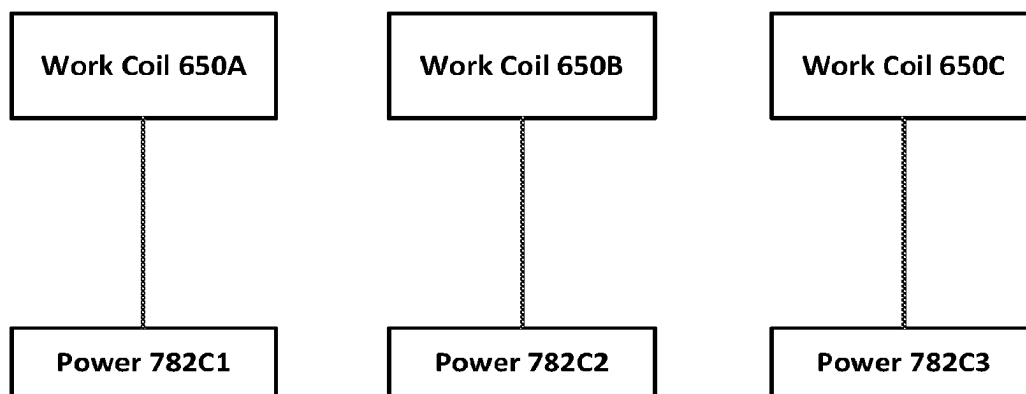


Figure 7C

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ELECTRICAL VAPOR PROVISION SYSTEM WITHOUT WIRED ELECTRICAL CONNECTION BETWEEN A CONTROL UNIT AND A CARTRIDGE

CROSS REFERENCE TO RELATED APPLICATION

The present application is a continuation application of U.S. patent application Ser. No. 17/456,446 filed Nov. 24, 2021, which is a continuation of application Ser. No. 15/739,037 filed Dec. 21, 2017, which is a National Phase entry of PCT Application No. PCT/GB2016/051766, filed Jun. 15, 2016, which claims priority from GB Application No. 1511359.0, filed Jun. 29, 2015, all of which are hereby fully incorporated herein by reference.

FIELD

The present disclosure relates to electronic vapor provision systems such as electronic nicotine delivery systems (e.g. e-cigarettes).

BACKGROUND

FIG. 1 is a schematic diagram of one example of a conventional e-cigarette 10. The e-cigarette has a generally cylindrical shape, extending along a longitudinal axis indicated by dashed line LA, and comprises two main components, namely a control unit 20 and a cartomizer 30. The cartomizer 30 includes an internal chamber containing a reservoir of liquid formulation including nicotine, a vaporizer (such as a heater), and a mouthpiece 35. The cartomizer 30 may further include a wick or similar facility to transport a small amount of liquid from the reservoir to the heater. The control unit 20 includes a re-chargeable battery to provide power to the e-cigarette 10 and a circuit board for generally controlling the e-cigarette 10. When the heater receives power from the battery, as controlled by the circuit board, the heater vaporizes the nicotine and this vapor (aerosol) is then inhaled by a user through the mouthpiece 35.

The control unit 20 and cartomizer 30 are detachable from one another by separating in a direction parallel to the longitudinal axis LA, as shown in FIG. 1, but are joined together when the device 10 is in use by a connection, indicated schematically in FIG. 1 as 25A and 25B, to provide mechanical and electrical connectivity between the control unit 20 and the cartomizer 30. The electrical connector on the control unit 20 that is used to connect to the cartomizer 30 also serves as a socket for connecting a charging device (not shown) when the control unit 20 is detached from the cartomizer 30. The cartomizer 30 may be detached from the control unit 20 and disposed of when the supply of nicotine is exhausted (and replaced with another cartomizer if so desired).

FIGS. 2 and 3 provide schematic diagrams of the control unit 20 and cartomizer 30 respectively of the e-cigarette 10 of FIG. 1. Note that various components and details, e.g. such as wiring and more complex shaping, have been omitted from FIGS. 2 and 3 for reasons of clarity. As shown in FIG. 2, the control unit 20 includes a battery or cell 210 for powering the e-cigarette 10, as well as a chip, such as a (micro)controller for controlling the e-cigarette 10. The controller 20 is attached to a small printed circuit board (PCB) 215 that also includes a sensor unit. If a user inhales on the mouthpiece 35, air is drawn into the e-cigarette 10 through one or more air inlet holes (not shown in FIGS. 1

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and 2). The sensor unit detects this airflow, and in response to such a detection, the controller provides power from the battery 210 to the heater in the cartomizer 30.

As shown in FIG. 3, the cartomizer 30 includes an air passage 161 extending along the central (longitudinal) axis LA of the cartomizer 30 from the mouthpiece 35 to the connector 25A for joining the cartomizer 30 to the control unit 20. A reservoir 170 of nicotine-containing liquid is provided around the air passage 161. This reservoir 170 may be implemented, for example, by providing cotton or foam soaked in the liquid. The cartomizer 30 also includes a heater 155 in the form of a coil for heating liquid from reservoir 170 to generate vapor to flow through air passage 161 and out through mouthpiece 35. The heater 155 is powered through lines 166 and 167, which are in turn connected to opposing polarities (positive and negative, or vice versa) of the battery 210 via connector 25A.

One end of the control unit provides a connector 25B for joining the control unit 20 to the connector 25A of the cartomizer 30. The connectors 25A and 25B provide mechanical and electrical connectivity between the control unit 20 and the cartomizer 30. The connector 25B includes two electrical terminals, an outer contact 240 and an inner contact 250, which are separated by insulator 260. The connector 25A likewise includes an inner electrode 175 and an outer electrode 171, separated by insulator 172. When the cartomizer 30 is connected to the control unit 20, the inner electrode 175 and the outer electrode 171 of the cartomizer 30 engage the inner contact 250 and the outer contact 240, respectively, of the control unit 20. The inner contact 250 is mounted on a coil spring 255 so that the inner electrode 175 pushes against the inner contact 250 to compress the coil spring 255, thereby helping to ensure good electrical contact when the cartomizer 30 is connected to the control unit 20.

The connector 25A is provided with two lugs or tabs 180A, 180B, which extend in opposite directions away from the longitudinal axis LA of the e-cigarette 10. These tabs 180A, 180B are used to provide a bayonet fitting for connecting the cartomizer 30 to the control unit 20. It will be appreciated that other embodiments may use a different form of connection between the control unit 20 and the cartomizer 30, such as a snap fit or a screw connection.

As mentioned above, the cartomizer 30 is generally disposed of once the liquid reservoir 170 has been depleted, and a new cartomizer is purchased and installed. In contrast, the control unit 20 is re-usable with a succession of cartomizers 30. Accordingly, it is particularly desirable to keep the cost of the cartomizer 30 relatively low. One approach to doing this has been to construct a three-part device, based on (i) a control unit, (ii) a vaporizer component, and (iii) a liquid reservoir. In this three-part device, only the final part, the liquid reservoir, is disposable, whereas the control unit and the vaporizer are both re-usable. However, having a three-part device can increase the complexity, both in terms of manufacture and user operation. Moreover, it can be difficult in such a three-part device to provide a wicking arrangement of the type shown in FIG. 3 to transport liquid from the reservoir to the heater.

Another approach is to make the cartomizer 30 re-fillable, so that it is no longer disposable. However, making a cartomizer 30 re-fillable brings potential problems, for example, a user may try to re-fill the cartomizer 30 with an inappropriate liquid (one not provided by the supplier of the e-cigarette). There is a risk that this inappropriate liquid may result in a low quality consumer experience, and/or may be potentially hazardous, whether by causing damage to the e-cigarette itself, or possibly by creating toxic vapors.

Accordingly, existing approaches for reducing the cost of a disposable component (or for avoiding the need for such a disposable component) have met with only limited success.

SUMMARY

The invention is defined in the appended claims.

Various embodiments of the disclosure provide an electronic vapor provision system having a longitudinal axis. The electronic vapor provision system comprises a control unit and a cartridge configured to engage with and disengage from one another substantially along said longitudinal axis. The cartridge includes a reservoir of liquid to be vaporized, and the control unit includes a battery and a tube portion disposed along said longitudinal axis. The system further comprises an induction heating assembly comprising a drive coil and a heater element. The heater element is located in the cartridge for vaporizing the liquid. The cartridge is configured to feed liquid from the reservoir onto the heater element for vaporization. The drive coil is located in the tube portion of the control unit and disposed about said longitudinal axis. The cartridge is located at least partly within the tube portion when engaged with the control unit, such that the heater element is within the drive coil.

The approach described herein is not restricted to specific embodiments such as set out below, but includes and contemplates any appropriate combinations of features presented herein. For example, an electronic vapor provision system may be provided in accordance with the approach described herein which includes any one or more of the various features described below as appropriate.

BRIEF DESCRIPTION OF THE DRAWINGS

Various embodiments of the disclosure will now be described in detail by way of example only with reference to the following drawings:

FIG. 1 is a schematic (exploded) diagram illustrating an example of a known e-cigarette.

FIG. 2 is a schematic diagram of the control unit of the e-cigarette of FIG. 1.

FIG. 3 is a schematic diagram of the cartomizer of the e-cigarette of FIG. 1.

FIG. 4 is a schematic diagram illustrating an e-cigarette in accordance with some embodiments of the disclosure, showing the control unit assembled with the cartridge (top), the control unit by itself (middle), and the cartridge by itself (bottom).

FIGS. 5 and 6 are schematic diagrams illustrating an e-cigarette in accordance with some other embodiments of the disclosure.

FIG. 7 is a schematic diagram of the control electronics for an e-cigarette such as shown in FIGS. 4, 5 and 6 in accordance with some embodiments of the disclosure.

FIGS. 7A, 7B and 7C are schematic diagrams of part of the control electronics for an e-cigarette such as shown in FIG. 6 in accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION

FIG. 4 is a schematic diagram illustrating an e-cigarette 410 in accordance with some embodiments of the disclosure (please note that the term e-cigarette is used herein interchangeably with other similar terms, such as electronic vapor provision system, electronic aerosol provision system,

etc.). The e-cigarette 410 includes a control unit 420 and a cartridge 430. FIG. 4 shows the control unit 420 assembled with the cartridge 430 (top), the control unit 420 by itself (middle), and the cartridge 430 by itself (bottom). Note that for clarity, various implementation details (e.g. such as internal wiring, etc.) are omitted.

As shown in FIG. 4, the e-cigarette 410 has a generally cylindrical shape with a central, longitudinal axis (denoted as LA, shown in dashed line). Note that the cross-section through the cylinder, i.e. in a plane perpendicular to the line LA, may be circular, elliptical, square, rectangular, hexagonal, or some other regular or irregular shape as desired. The general profile and shape (form factor) of the e-cigarette 410 is (or can be arranged to be) the same or broadly similar to that of e-cigarette 10 as shown in FIGS. 1-3. This consistency can be helpful for various reasons, such as potential to share components and packaging, user acceptance and ease of operation, brand recognition, etc.

The mouthpiece 435 is located at one end of the cartridge 430, while the opposite end of the e-cigarette 410 (with respect to the longitudinal axis) is denoted as the tip end 424. The end of the cartridge 430 which is longitudinally opposite to the mouthpiece 435 is denoted by reference numeral 431, while the end of the control unit 420 which is longitudinally opposite to the tip end 424 is denoted by reference numeral 421.

The cartridge 430 is able to engage with and disengage from the control unit 420 by movement along the longitudinal axis LA. More particularly, the end 431 of the cartridge 430 is able to engage with, and disengage from, the end 421 of the control unit 420. Accordingly, ends 421 and 431 will be referred to as the engagement end 421 of the control unit 420 and the engagement end 431 of the cartridge 430, respectively.

The control unit 420 includes a battery 411 and a circuit board 415 to provide control functionality for the e-cigarette 410, e.g. by provision of a controller, processor, Application Specific Integrated Circuit (ASIC) or similar form of control chip. The battery 411 is typically cylindrical in shape, and has a central axis that lies along, or at least close to, the longitudinal axis LA of the e-cigarette 410. In FIG. 4, the circuit board 415 is shown longitudinally spaced from the battery 411, in the opposite direction to the cartridge 430. However, the skilled person will be aware of various other locations for the circuit board 415, for example, it may be at the opposite end of the battery 411. A further possibility is that the circuit board 415 lies along the side of the battery 411—for example, with the e-cigarette 410 having a rectangular cross-section, the circuit board 415 located adjacent one outer wall of the e-cigarette 410, and the battery 411 then slightly offset towards the opposite outer wall of the e-cigarette 410. Note also that the functionality provided by the circuit board 415 (as described in more detail below) may be split across multiple circuit boards and/or across devices which are not mounted to a PCB, and these additional devices and/or PCBs can be located as appropriate within the e-cigarette 410.

The battery or cell 411 is generally re-chargeable, and one or more re-charging mechanisms may be supported. For example, a charging connection (not shown in FIG. 4) may be provided at the tip end 424, and/or the engagement end 421 of the control unit 420, and/or along the side of the e-cigarette 410. Moreover, the e-cigarette 410 may support induction re-charging of battery 411, in addition to (or instead of) re-charging via one or more re-charging connections or sockets.

The control unit 420 includes a tube portion 440, which extends along the longitudinal axis LA away from the engagement end 421 of the control unit 420. The tube portion 440 is defined on the outside by outer wall 442, which may generally be part of the overall outer wall or housing of the control unit 420, and on the inside by inner wall 444. A cavity 426 is formed by inner wall 444 of the tube portion and the engagement end 421 of the control unit 420. This cavity 426 is able to receive and accommodate at least part of a cartridge 430 as it engages with the control unit 420 (as shown in the top drawing of FIG. 4).

The inner wall 444 and the outer wall 442 of the tube portion define an annular space which is formed around the longitudinal axis LA. A drive coil 450 is located within this annular space, with the central axis of the drive coil 450 being substantially aligned with the longitudinal axis LA of the e-cigarette 410. A drive coil 450 may sometimes also be referred to as a work coil. The drive coil 450 is electrically connected to the battery 411 and circuit board 415, which provide power and control to the drive coil 450, so that in operation, the drive coil 450 is able to provide induction heating to the cartridge 430.

The cartridge 430 includes a reservoir 470 containing liquid formulation (typically including nicotine). The reservoir 470 comprises a substantially annular region of the cartridge 430, formed between an outer wall 476 of the cartridge 430, and an inner tube or wall 472 of the cartridge 430, both of which are substantially aligned with the longitudinal axis LA of the e-cigarette 410. The liquid formulation may be held free within the reservoir 470, or alternatively the reservoir 470 may be incorporated in some structure or material, e.g. sponge, to help retain the liquid within the reservoir 470.

The outer wall 476 has a portion 476A of reduced cross-section of the cartridge 430. This allows this portion 476A of the reduced cross-section of the cartridge 430 to be received into the cavity 426 in the control unit 420 in order to engage the cartridge 430 with the control unit 420. The remainder of the outer wall 476 has a greater cross-section in order to provide increased space within the reservoir 470, and also to provide a continuous outer surface for the e-cigarette 410—i.e. outer wall 476 of the cartridge is substantially flush with the outer wall 442 of the tube portion 440 of the control unit 420. However, it will be appreciated that other implementations of the e-cigarette 410 may have a more complex/structured outer surface (compared with the smooth outer surface shown in FIG. 4).

The inside of the inner tube 472 defines a passageway 461 which extends, in a direction of airflow, from air inlet 461A (located at the engagement end 431 of the cartridge 430 that engages the control unit 420) through to air outlet 461B, which is provided by the mouthpiece 435. Located within the central passageway 461, and hence within the airflow through the cartridge 430, are heater element 455 and wick 454. As can be seen in FIG. 4, the heater element 455 is located approximately in the center of the drive coil 450. In particular, the location of the heater element 455 along the longitudinal axis LA can be controlled by having the step at the start of the portion 476A of reduced cross-section for the cartridge 430 abut against the end (nearest the mouthpiece 435) of the tube portion 440 of the control unit 420 (as shown in the top diagram of FIG. 4).

The heater element 455 is made of a metallic material so as to permit use as a susceptor (or workpiece) in an induction heating assembly. More particularly, the induction heating assembly comprises the drive coil 450, which produces a magnetic field having high frequency variations (when suit-

ably powered and controlled by the battery 411 and controller on PCB 415). This magnetic field is strongest in the center of the drive coil 450, i.e. within cavity 426, where the heater element 455 is located. The changing magnetic field induces eddy currents in the heater element 455 (which is capable of being conductively heated), thereby causing resistive heating within the heater element 455. Note that the high frequency of the variations in magnetic field causes the eddy currents to be confined to the surface of the heater element 455 (via the skin effect), thereby increasing the effective resistance of the heater element 455, and hence the resulting heating effect.

Furthermore, the heater element 455 is generally selected to be a magnetic material having a high permeability, such as (ferrous) steel (rather than just a conductive material). In this case, the resistive losses due to eddy currents are supplemented by magnetic hysteresis losses (caused by repeated flipping of magnetic domains) to provide more efficient transfer of power from the drive coil 450 to the heater element 455.

The heater element 455 is at least partly surrounded by wick 454. Wick 454 serves to transport liquid from the reservoir 470 onto the heater element 455 for vaporization. The wick 454 may be made of any suitable material, for example, a heat-resistant, fibrous material and typically extends from the passageway 461 through holes in the inner tube 472 to gain access into the reservoir 470. The wick 454 is arranged to supply liquid to the heater element 455 in a controlled manner, in that the wick 454 prevents the liquid leaking freely from the reservoir 470 into passageway 461 (this liquid retention may also be assisted by having a suitable material within the reservoir 470 itself). Instead, the wick 454 retains the liquid within the reservoir 470, and on the wick 454 itself, until the heater element 455 is activated, whereupon the liquid held by the wick 454 is vaporized into the airflow, and hence travels along passageway 461 for exit via mouthpiece 435. The wick 454 then draws further liquid into itself from the reservoir 470, and the process repeats with subsequent vaporizations (and inhalations) until the cartridge 430 is depleted.

Although the wick 454 is shown in FIG. 4 as separate from (albeit encompassing) the heater element 455, in some implementations, the heater element 455 and wick 454 may be combined together into a single component, such as a heater element 455 made of a porous, fibrous steel material which can also act as a wick 454 (as well as a heater). In addition, although the wick 454 is shown in FIG. 4 as supporting the heater element 455, in other embodiments, the heater element 455 may be provided with separate supports, for example, by being mounted to the inner tube 472 (instead of or in addition to being supported by the heater element 455).

The heater element 455 may be substantially planar, and perpendicular to the central axis of the drive coil 450 and the longitudinal axis LA of the e-cigarette 410, since induction primarily occurs in this plane. Although FIG. 4 shows the heater element 455 and wick 454 extending across the full diameter of the inner tube 472, typically the heater element 455 and wick 454 will not cover the whole cross-section of the air passage-way 461. Instead, space is typically provided to allow air to flow through the inner tube 472 from air inlet 461A and around heater element 455 and wick 454 to pick up the vapor produced by the heater. For example, when viewed along the longitudinal axis LA, the heater element 455 and wick 454 may have an "O" configuration with a central hole (not shown in FIG. 4) to allow for airflow along the passageway 461. Many other configurations are

possible, such as the heater having a “Y” or “X” configuration. (Note that in such implementations, the arms of the “Y” or “X” would be relatively broad to provide better induction.)

Although FIG. 4 shows the engagement end 431 of the cartridge 430 as covering the air inlet 461A, this end of the cartomizer may be provided with one or more holes (not shown in FIG. 4) to allow the desired air intake to be drawn into passageway 461. Note also that in the configuration shown in FIG. 4, there is a slight gap 422 between the engagement end 431 of the cartridge 430 and the corresponding engagement end 421 of the control unit 420. Air can be drawn from this gap 422 through air inlet 461A.

The e-cigarette 410 may provide one or more routes to allow air to initially enter the gap 422. For example, there may be sufficient spacing between the outer wall 476 of the cartridge 430 and the inner wall 444 of tube portion 440 to allow air to travel into gap 422. Such spacing may arise naturally if the cartridge 430 is not a tight fit into the cavity 426. Alternatively one or more air channels may be provided as slight grooves along one or both of these walls 476, 444 to support this airflow. Another possibility is for the housing of the control unit 420 to be provided with one or more holes, firstly to allow air to be drawn into the control unit 420, and then to pass from the control unit 420 into gap 422. For example, the holes for air intake into the control unit 420 might be positioned as indicated in FIG. 4 by arrows 428A and 428B, and engagement end 421 of control unit 420 might be provided with one or more holes (not shown in FIG. 4) for the air to pass out from the control unit 420 into gap 422 (and from there into the cartridge 430). In other implementations, gap 422 may be omitted, and the airflow may, for example, pass directly from the control unit 420 through the air inlet 461A into the cartridge 430.

The e-cigarette 410 may be provided with one or more activation mechanisms for the induction heater assembly, i.e. to trigger operation of the drive coil 450 to heat the heater element 455. One possible activation mechanism is to provide a button 429 on the control unit, which a user may press to activate the heater element 455. This button may be a mechanical device, a touch sensitive pad, a sliding control, etc. The heater element 455 may stay activated for as long as the user continues to press or otherwise positively actuate the button 429, subject to a maximum activation time appropriate to a single puff of the e-cigarette (typically a few seconds). If this maximum activation time is reached, the controller may automatically de-activate the induction heater to prevent over-heating. The controller may also enforce a minimum interval (again, typically for a few seconds) between successive activations.

The induction heater assembly may also be activated by airflow caused by a user inhalation. In particular, the control unit 420 may be provided with an airflow sensor for detecting an airflow (or pressure drop) caused by an inhalation. The airflow sensor is then able to notify the controller of this detection, and the induction heater is activated accordingly. The induction heater may remain activated for as long as the airflow continues to be detected, subject again to a maximum activation time as above (and typically also a minimum interval between puffs).

Airflow actuation of the heater element 455 may be used instead of providing button 429 (which could therefore be omitted), or alternatively the e-cigarette 410 may require dual activation in order to operate—i.e. both the detection of airflow and the pressing of button 429. This requirement for dual activation can help to provide a safeguard against unintended activation of the e-cigarette 410.

It will be appreciated that the use of an airflow sensor generally involves an airflow passing through the control unit 420 upon inhalation, which is amenable to detection (even if this airflow only provides part of the airflow that the user ultimately inhales). If no such airflow passes through the control unit 420 upon inhalation, then button 429 may be used for activation, although it might also be possible to provide an airflow sensor to detect an airflow passing across a surface of (rather than through) the control unit 420.

There are various ways in which the cartridge 430 may be retained within the control unit 420. For example, the inner wall 444 of the tube portion 440 of the control unit 420 and the outer wall 476 of the portion 476A of reduced cross-section of the cartridge 430 may each be provided with a screw thread (not shown in FIG. 4) for mutual engagement. Other forms of mechanical engagement, such as a snap fit, a latching mechanism (perhaps with a release button or similar) may also be used. Furthermore, the control unit 420 may be provided with additional components to provide a fastening mechanism, such as described below.

In general terms, the attachment of the cartridge 430 to the control unit 420 for the e-cigarette 410 of FIG. 4 is simpler than in the case of the e-cigarette 10 shown in FIGS. 1-3. In particular, the use of induction heating for e-cigarette 410 allows the connection between the cartridge 430 and the control unit 420 to be mechanical only, rather than also having to provide an electrical connection with wiring to a resistive heater. Consequently, the mechanical connection may be implemented, if so desired, by using an appropriate plastic molding for the housing of the cartridge 430 and the control unit 420; in contrast, in the e-cigarette 10 of FIGS. 1-3, the housings of the cartomizer 30 and the control unit 20 have to be somehow bonded to a metal connector. Furthermore, the connector of the e-cigarette 10 of FIGS. 1-3 has to be made in a relatively precise manner to ensure a reliable, low contact resistance, electrical connection between the control unit 20 and the cartomizer 30. In contrast, the manufacturing tolerances for the purely mechanical connection between the cartridge 430 and the control unit 420 of e-cigarette 410 are generally greater. These factors all help to simplify the production of the cartridge 430 and thereby to reduce the cost of this disposable (consumable) component.

Furthermore, conventional resistive heating often utilizes a metallic heating coil surrounding a fibrous wick, however, it is relatively difficult to automate the manufacture of such a structure. In contrast, an inductive heater element 455 is typically based on some form of metallic disk (or other substantially planar component), which is an easier structure to integrate into an automated manufacturing process. This again helps to reduce the cost of production for the disposable cartridge 430.

Another benefit of inductive heating is that conventional e-cigarettes may use solder to bond power supply wires to a resistive heater coil. However, there is some concern that heat from the coil during operation of such an e-cigarette might volatilize undesirable components from the solder, which would then be inhaled by a user. In contrast, there are no wires to bond to the heater element 455 (which is capable of being inductively heated), and hence the use of solder can be avoided within the cartridge 430. Also, a resistive heater coil as in a conventional e-cigarette generally comprises a wire of relatively small diameter (to increase the resistance and hence the heating effect). However, such a thin wire is relatively delicate and so may be susceptible to damage, whether through some mechanical mistreatment and/or potentially by local overheating and then melting. In con-

trast, a disk-shaped heater element **455** as used for induction heating is generally more robust against such damage.

FIGS. **5** and **6** are schematic diagrams illustrating an e-cigarette in accordance with some other embodiments of the disclosure. To avoid repetition, aspects of FIGS. **5** and **6** that are generally the same as shown in FIG. **4** will not be described again, except where relevant to explain the particular features of FIGS. **5** and **6**. Note also that reference numbers having the same last two digits typically denote the same or similar (or otherwise corresponding) components across FIGS. **4** to **6** (with the first digit in the reference number corresponding to the Figure containing that reference number).

In the e-cigarette **510** shown in FIG. **5**, the control unit **520** is broadly similar to the control unit **420** shown in FIG. **4**, however, the internal structure of the cartridge **530** is somewhat different from the internal structure of the cartridge **430** shown in FIG. **4**. Thus rather than having a central airflow passage, as for e-cigarette **410** of FIG. **4**, in which the liquid reservoir **470** surrounds the central airflow passage **461**, in the e-cigarette **510** of FIG. **5**, the air passageway **561** is offset from the central, longitudinal axis (LA) of the cartridge **530**. In particular, the cartridge **530** contains an inner wall **572** that separates the internal space of the cartridge **530** into two portions. A first portion, defined by inner wall **572** and one part of outer wall **576**, provides a chamber for holding the reservoir **570** of liquid formulation. A second portion, defined by inner wall **572** and an opposing part of outer wall **576**, defines the air passage way **561** through the e-cigarette **510**.

In addition, the e-cigarette **510** does not have a wick, but rather relies upon a porous heater element **555** to act both as the heating element (susceptor) and the wick to control the flow of liquid out of the reservoir **570**. The porous heater element **555** may be made, for example, of a material formed from sintering or otherwise bonding together steel fibers.

The heater element **555** is located at the end of the reservoir **570** opposite to the mouthpiece **535** of the cartridge **530**, and may form some or all of the wall of the reservoir chamber at this end. One face of the heater element is in contact with the liquid in the reservoir **570**, while the opposite face of the heater element **555** is exposed to an airflow region **538** which can be considered as part of air passageway **561**. In particular, this airflow region **538** is located between the heater element **555** and the engagement end **531** of the cartridge **530**.

When a user inhales on mouthpiece **535**, air is drawn into the region **538** through the engagement end **531** of the cartridge **530** from gap **522** (in a similar manner to that described for the e-cigarette **410** of FIG. **4**). In response to the airflow (and/or in response to the user pressing button **529**), the coil **550** is activated to supply power to heater **555**, which therefore produces a vapor from the liquid in reservoir **570**. This vapor is then drawn into the airflow caused by the inhalation, and travels along the air passageway **561** (as indicated by the arrows) and out through mouthpiece **535**.

In the e-cigarette **610** shown in FIG. **6**, the control unit **620** is broadly similar to the control unit **420** shown in FIG. **4**, but now accommodates two (smaller) cartridges **630A**, and **630B**. Each of these cartridges **630A**, **630B** is analogous in structure to the portion **476A** of reduced cross-section of the cartridge **430** in FIG. **4**. However, the longitudinal extent of each of the cartridges **630A** and **630B** is only half that of the portion **476A** of reduced cross-section of the cartridge **430** in FIG. **4**, thereby allowing two cartridges to be contained within the region in e-cigarette **610** corresponding to cavity **426** in e-cigarette **410**, as shown in FIG. **4**. In

addition, the engagement end **621** of the control unit **620** may be provided, for example, with one or more struts or tabs (not shown in FIG. **6**) that maintain cartridges **630A**, **630B** in the position shown in FIG. **6** (rather than closing the gap **622**).

In the e-cigarette **610**, the mouthpiece **635** may be regarded as part of the control unit **620**. In particular, the mouthpiece **635** may be provided as a removable cap or lid, which can screw or clip onto and off the remainder of the control unit **620** (or any other appropriate fastening mechanism can be used). The mouthpiece **635** is removed from the rest of the control unit **620** to insert a new cartridge or to remove an old cartridge, and then fixed back onto the control unit for use of the e-cigarette **610**.

The operation of the individual cartridges **630A**, **630B** in e-cigarette **610** is similar to the operation of cartridge **430** in e-cigarette **410**, in that each cartridge includes a wick **654A**, **654B** extending into the respective reservoir **670A**, **670B**. In addition, each cartridge **630A**, **630B** includes a heating element, **655A**, **655B**, accommodated in a respective wick, **654A**, **654B**, and may be energized by a respective coil **650A**, **650B** provided in the control unit **620**. The heaters **655A**, **655B** vaporize liquid into a common passageway **661** that passes through both cartridges **630A**, **630B** and out through mouthpiece **635**.

The different cartridges **630A**, **630B** may be used, for example, to provide different flavors for the e-cigarette **610**. In addition, although the e-cigarette **610** is shown as accommodating two cartridges **630A**, **630B**, it will be appreciated that some devices may accommodate a larger number of cartridges. Furthermore, although cartridges **630A** and **630B** are the same size as one another, some devices may accommodate cartridges of differing size. For example, an e-cigarette may accommodate one larger cartridge having a nicotine-based liquid, and one or more small cartridges to provide flavor or other additives as desired.

In some cases, the e-cigarette **610** may be able to accommodate (and operate with) a variable number of cartridges. For example, there may be a spring or other resilient device mounted on control unit engagement end **621**, which tries to extend along the longitudinal axis towards the mouthpiece **635**. If one of the cartridges **630A**, **630B** shown in FIG. **6** is removed, this spring would therefore help to ensure that the remaining cartridge(s) would be held firmly against the mouthpiece **635** for reliable operation.

If an e-cigarette has multiple cartridges, one option is that these are all activated by a single coil that spans the longitudinal extent of all the cartridges. Alternatively, there may be an individual coil **650A**, **650B** for each respective cartridge **630A**, **630B**, as illustrated in FIG. **6**. A further possibility is that different portions of a single coil may be selectively energized to mimic (emulate) the presence of multiple coils.

If an e-cigarette does have multiple coils for respective cartridges (whether really separate coils, or emulated by different sections of a single larger coil), then activation of the e-cigarette (such as by detecting airflow from an inhalation and/or by a user pressing a button) may energize all coils. The e-cigarettes **410**, **510**, **610**, however, support selective activation of the multiple coils, whereby a user can choose or specify which coil(s) to activate. For example, e-cigarette **610** may have a mode or user setting in which in response to an activation, only coil **650A** is energized, but not coil **650B**. This would then produce a vapor based on the liquid formulation in coil **650A**, but not coil **650B**. This would allow a user greater flexibility in the operation of e-cigarette **610**, in terms of the vapor provided for any given

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inhalation (but without a user having to physically remove or insert different cartridges just for that particular inhalation).

It will be appreciated that the various implementations of e-cigarette **410**, **510** and **610** shown in FIGS. 4-6 are provided as examples only, and are not intended to be exhaustive. For example, the cartridge design shown in FIG. 5 might be incorporated into a multiple cartridge device such as shown in FIG. 6. The skilled person will be aware of many other variations that can be achieved, for example, by mixing and matching different features from different implementations, and more generally by adding, replacing and/or removing features as appropriate.

FIG. 7 is a schematic diagram of the main electronic components of the e-cigarettes **410**, **510**, **610** of FIGS. 4-6 in accordance with some embodiments of the disclosure. The heater element **455**, which is located in the cartridge **430**, may comprise any suitable structure, or combination of structures, for inductive heating. The remaining elements shown in FIG. 7 are located in the control unit **420**. It will be appreciated that since the control unit **420** is a re-usable device (in contrast to the cartridge **430** which is a disposable or consumable), it is acceptable to incur one-off costs in relation to production of the control unit **420** which would not be acceptable as repeat costs in relation to the production of the cartridge **430**. The components of the control unit **420** may be mounted on circuit board **415**, or may be separately accommodated in the control unit **420** to operate in conjunction with the circuit board **415** (if provided), but without being physically mounted on the circuit board **415** itself.

As shown in FIG. 7, the control unit **420** includes a re-chargeable battery **411**, which is linked to a re-charge connector or socket **725**, such as a micro-USB interface. This connector **725** supports re-charging of battery **411**. Alternatively, or additionally, the control unit **420** may also support re-charging of battery **411** by a wireless connection (such as by induction charging).

The control unit **420** further includes a controller **715** (such as a processor or application specific integrated circuit, ASIC), which is linked to a pressure or airflow sensor **716**. The controller **715** may activate the induction heating, as discussed in more detail below, in response to the sensor **716** detecting an airflow. In addition, the control unit **420** further includes a button **429**, which may also be used to activate the induction heating, as described above.

FIG. 7 also shows a comms/user interface **718** for the e-cigarette. This may comprise one or more facilities according to the particular implementation. For example, the comms/user interface **718** may include one or more lights and/or a speaker to provide output to the user, for example to indicate a malfunction, battery charge status, etc. The comms/user interface **718** may also support wireless communications, such as Bluetooth or near field communications (NFC), with an external device, such as a smartphone, laptop, computer, notebook, tablet, etc. The e-cigarette may utilize this comms/user interface **718** to output information such as device status, usage statistics, etc., to the external device, for ready access by a user. The comms/user interface **718** may also be utilized to allow the e-cigarette to receive instructions, such as configuration settings entered by the user into the external device. For example, the comms/user interface **718** and controller **715** may be utilized to instruct the e-cigarette to selectively activate different coils **650A**, **650B** (or portions thereof), as described above. In some cases, the comms/user interface **718** may use the drivesoil **450** to act as an antenna for wireless communications.

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The controller **715** may be implemented using one or more chips as appropriate. The operations of the controller **715** are generally controlled at least in part by software programs running on the controller **715**. Such software programs may be stored in non-volatile memory, such as Read Only Memory (ROM), which can be integrated into the controller **715** itself, or provided as a separate component (not shown). The controller **715** may access the ROM to load and execute individual software programs as and when required.

The controller **715** controls the inductive heating of the e-cigarette by determining when the device is or is not properly activated—for example, whether an inhalation has been detected, and whether the maximum time period for an inhalation has not yet been exceeded. If the controller **715** determines that the e-cigarette is to be activated for vaping, the controller **715** arranges for the battery **411** to supply power to the inverter **712**. The inverter **712** is configured to convert the DC output from the battery **411** into an alternating current signal, typically of relatively high frequency—e.g. 1 MHz (although other frequencies, such as 5 kHz, 20 kHz, 80 KHz, or 300 kHz, or any range defined by two such values, may be used instead). This Alternating Current (AC) signal is then passed from the inverter to the drive coil **450**, via suitable impedance matching (not shown in FIG. 7) if so required.

The drive coil **450** may be integrated into some form of resonant circuit, such as by combining in parallel with a capacitor (not shown in FIG. 7), with the output of the inverter **712** tuned to the resonant frequency of this resonant circuit. This resonance causes a relatively high current to be generated in drive coil **450**, which in turn produces a relatively high magnetic field in heater element **455**, thereby causing rapid and effective heating of the heater element **455** to produce the desired vapor or aerosol output.

FIG. 7A illustrates part of the control electronics for an e-cigarette **610** having multiple coils in accordance with some implementations (while omitting for clarity aspects of the control electronics not directly related to the multiple coils). FIG. 7A shows a power source **782A** (typically corresponding to the battery **411** and inverter **712** of FIG. 7), a switch configuration **781A**, and the two work coils **650A**, **650B**, each associated with a respective heater element **655A**, **655B** as shown in FIG. 6 (but not included in FIG. 7A). The switch configuration has three outputs denoted A, B and C in FIG. 7A. It is also assumed that there is a current path between the two work coils **650A**, **650B**.

In order to operate the induction heating assembly, two out of three of these outputs A, B, C are closed (to permit current flow), while the remaining output stays open (to prevent current flow). Closing outputs A and C activates both coils, and hence both heater elements **655A**, **655B**; closing A and B selectively activates just work coil **650A**; and closing B and C activates just work coil **650B**.

Although it is possible to treat work coils **650A** and **650B** just as a single overall coil (which is either on or off together), the ability to selectively energize either or both of work coils **650A** and **650B**, such as provided by the implementation of FIG. 7, has a number of advantages, including:

- choosing the vapor components (e.g. flavorants) for a given puff. Thus activating just work coil **650A** produces vapor just from reservoir **670A**; activating just work coil **650B** produces vapor just from reservoir **670B**; and activating both work coils **650A**, **650B** produces a combination of vapor from both reservoirs **670A**, **670B**.

- b) controlling the amount of vapor for a given puff. For example, if reservoir 670A and reservoir 670B in fact contain the same liquid, then activating both work coils 650A, 650B can be used to produce a stronger (higher vapor level) puff compared to activating just one work coil by itself
- c) prolonging battery (charge) lifetime. As already discussed, it may be possible to operate the e-cigarette 610 of FIG. 6 when it contains just a single cartridge, e.g. 630B (rather than also including cartridge 630A). In this case, it is more efficient just to energize the work coil 650B corresponding to cartridge 630B, which is then used to vaporize liquid from reservoir 670B. In contrast, if the work coil 650A corresponding to the (missing) cartridge 630A is not energized (because this cartridge and the associated heater element 650A are missing from e-cigarette 610), then this saves power consumption without reducing vapor output.

Although the e-cigarette 610 of FIG. 6 has a separate heater element 655A, 655B for each respective work coil 650A, 650B, in some implementations, different work coils may energize different portions of a single (larger) work-piece or susceptor. Accordingly, in such an e-cigarette 610, the different heater elements 655A, 655B may represent different portions of the larger susceptor, which is shared across different work coils. Additionally (or alternatively), the multiple work coils 650A, 650B may represent different portions of a single overall drive coil, individual portions of which can be selectively energized, as discussed above in relation to FIG. 7A.

If the multiple heater elements are being used to control the amount of vapor for a given puff, then having a larger number of heater elements, e.g. more than the two shown in FIG. 7A, will give a better granularity of control. It will also be appreciated that the vapor amount can be increased by supplying more electrical power to each work coil for energizing the corresponding heater element, however, there are limits on the practicality of this. For example, providing too much power may result in a very high temperature for the heater element, which may alter the chemical constitution of the vapor, as well as presenting a potential safety concern.

FIG. 7B shows another implementation for supporting selectivity across multiple work coils 650A, 650B. Thus in FIG. 7B, it is assumed that the work coils 650A, 650B are not electrically connected to one another, but rather each work coil 650A, 650B is individually (separately) linked to the power source 782B via a pair of independent connections through switch configuration 781B. In particular, work coil 650A is linked to power source 782B via switch connections A1 and A2, and work coil 650B is linked to power source 782B via switch connections B1 and B2. This configuration of FIG. 7B offers similar advantages to those discussed above in relation to FIG. 7A. In addition, the architecture of FIG. 7B may also be readily scaled up to work with more than two work coils.

FIG. 7C shows another implementation for supporting selectivity across multiple work coils, in this case three work coils denoted 650A, 650B and 650C. Each work coil 650A, 650B, 650C is directly connected to a respective power supply 782C1, 782C2 and 782C3. The configuration of FIG. 7 may support the selective energization of any single work coil, 650A, 650B, 650C, or of any pair of work coils at the same time, or of all three work coils at the same time.

In the configuration of FIG. 7C, at least some portions of the power supply 782 may be replicated for each of the different work coils 650. For example, each power supply

782C1, 782C2, 782C3 may include its own inverter, but they may share a single, ultimate power source, such as battery 411. In this case, the battery 411 may be connected to the inverters via a switch configuration analogous to that shown in FIG. 7B (but for Direct Current (DC) rather than AC current). Alternatively, each respective power line from a power supply 782 to a work coil 650 may be provided with its own individual switch, which can be closed to activate the work coil (or opened to prevent such activation). In this arrangement, the collection of these individual switches across the different lines can be regarded as another form of switch configuration.

There are various ways in which the switching of FIGS. 7A-7C may be managed or controlled. In some cases, the user may operate a mechanical or physical switch that directly sets the switch configuration. For example, e-cigarette 610 may include a switch (not shown in FIG. 6) on the outer housing, whereby cartridge 630A can be activated in one setting, and cartridge 630B can be activated in another setting. A further setting of the switch may allow activation of both cartridges together. Alternatively, the control unit 620 may have a separate button associated with each cartridge, and the user holds down the button for the desired cartridge (or potentially both buttons if both cartridges should be activated). Another possibility is that a button or other input device on the e-cigarette 610 may be used to select a stronger puff (and result in switching on both or all work coils). Such a button may also be used to select the addition of a flavor, and the switching might operate a work coil associated with that flavor—typically in addition to a work coil for the base liquid containing nicotine. The skilled person will be aware of other possible implementations of such switching.

In some e-cigarettes, rather than direct (e.g. mechanical or physical) control of the switch configuration, the user may set the switch configuration via the comms/user interface 718 shown in FIG. 7 (or any other similar facility). For example, this comms/user interface 718 may allow a user to specify the use of different flavors or cartridges (and/or different strength levels), and the controller 715 can then set the switch configuration 781 according to this user input.

A further possibility is that the switch configuration may be set automatically. For example, e-cigarette 610 may prevent work coil 650A from being activated if a cartridge is not present in the illustrated location of cartridge 630A. In other words, if no such cartridge is present, then the work coil 650A may not be activated (thereby saving power, etc.).

There are various mechanisms available for detecting whether or not a cartridge is present. For example, the control unit 620 may be provided with a switch which is mechanically operated by inserting a cartridge into the relevant position. If there is no cartridge in position, then the switch is set so that the corresponding work coil is not powered. Another approach would be for the control unit to have some optical or electrical facility for detecting whether or not a cartridge is inserted into a given position.

Note that in some devices, once a cartridge has been detected as in position, then the corresponding work coil is always available for activation—e.g. it is always activated in response to a puff (inhalation) detection. In other devices that support both automatic and user-controlled switch configuration, even if a cartridge has been detected as in position, a user setting (or such-like, as discussed above) may then determine whether or not the cartridge is available for activation on any given puff.

Although the control electronics of FIGS. 7A-7C have been described in connection with the use of multiple

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cartridges, such as shown in FIG. 6, they may also be utilized in respect of a single cartridge that has multiple heater elements. In other words, the control electronics is able to selectively energize one or more of these multiple heater elements within the single cartridge. Such an approach may still offer the benefits discussed above. For example, if the cartridge contains multiple heater elements, but just a single, shared reservoir, or multiple heater elements, each with its own respective reservoir, but all reservoirs containing the same liquid, then energizing more or fewer heater elements provides a way for a user to increase or decrease the amount of vapor provided with a single puff. Similarly, if a single cartridge contains multiple heater elements, each with its own respective reservoir containing a particular liquid, then energizing different heater elements (or combinations thereof) provides a way for a user to selectively consume vapors for different liquids (or combinations thereof).

In some e-cigarettes, the various work coils and their respective heater elements (whether implemented as separate work coils and/or heater elements, or as portions of a larger drive coil and/or susceptor) may all be substantially the same as one another, to provide a homogeneous configuration. Alternatively, a heterogeneous configuration may be utilized. For example, with reference to e-cigarette 610 as shown in FIG. 6, one cartridge 630A may be arranged to heat to a lower temperature than the other cartridge 630B, and/or to provide a lower output of vapor (by providing less heating power). Thus if one cartridge 630A contains the main liquid formulation containing nicotine, while the other cartridge 630B contains a flavorant, it may be desirable for cartridge 630A to output more vapor than cartridge 630B. Also, the operating temperature of each heater element 655 may be arranged according to the liquid(s) to be vaporized. For example, the operating temperature should be high enough to vaporize the relevant liquid(s) of a particular cartridge, but typically not so high as to chemically break down (disassociate) such liquids.

There are various ways of providing different operating characteristics (such as temperature) for different combinations of work coils and heater elements, and thereby produce a heterogeneous configuration as discussed above. For example, the physical parameters of the work coils and/or heater elements may be varied as appropriate—e.g. different sizes, geometry, materials, number of coil turns, etc. Additionally (or alternatively), the operating parameters of the work coils and/or heater elements may be varied, such as by having different AC frequencies and/or different supply currents for the work coils.

In order to address various issues and advance the art, this disclosure shows by way of illustration various embodiments in which the claimed invention(s) may be practiced. The advantages and features of the disclosure are of a representative sample of embodiments only, and are not exhaustive and/or exclusive. They are presented only to assist in understanding and to teach the claimed invention(s). It is to be understood that advantages, embodiments, examples, functions, features, structures, and/or other aspects of the disclosure are not to be considered limitations on the disclosure as defined by the claims or limitations on equivalents to the claims, and that other embodiments may be utilized and modifications may be made without departing from the scope of the claims. Various embodiments may suitably comprise, consist of, or consist essentially of, various combinations of the disclosed elements, components, features, parts, steps, means, etc other than those

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specifically described herein. The disclosure may include other inventions not presently claimed, but which may be claimed in future.

The invention claimed is:

1. An electronic vapor provision system having a longitudinal axis and comprising:

a control unit including a control unit engagement end and a cartridge including a cartridge engagement end configured to engage with and disengage from one another substantially along said longitudinal axis, and the control unit including a battery and a tube portion disposed along said longitudinal axis, wherein the tube portion comprises an inner wall and an outer wall; and

an induction heating assembly comprising a drive coil and a heater element, wherein the heater element is located in the cartridge, the drive coil being located in the tube portion of the control unit, between the inner wall and the outer wall of the tube portion, and disposed about said longitudinal axis, wherein the cartridge is located at least partly within the inner wall of the tube portion when engaged with said control unit, such that the heater element is within the drive coil,

wherein the control unit is configured such that when the cartridge is engaged with the control unit, a gap is formed between the cartridge engagement end and the control unit engagement end, whereby, in use, air is able to be drawn from the gap through an air inlet of the cartridge provided in the cartridge engagement end; and

wherein the control unit is provided with one or more holes in a housing of the control unit to allow, in use, air to be drawn into the control unit and into the gap.

2. The electronic vapor provision system of claim 1, wherein the cartridge comprises a mouthpiece.

3. The electronic vapor provision system of claim 1, wherein the heater element is substantially planar and substantially perpendicular to the longitudinal axis.

4. The electronic vapor provision system of claim 1, wherein the heater element is made of a material having a steel.

5. The electronic vapor provision system of claim 1, wherein the heater element is made of a material having a ferrous steel.

6. The electronic vapor provision system of claim 1, wherein there is no wired electrical connection between the control unit and the cartridge.

7. The electronic vapor provision system of claim 1, wherein the cartridge does not have any external metal connector for electrical connection between the cartridge and the control unit.

8. The electronic vapor provision system of claim 1, wherein the control unit accommodates multiple cartridges.

9. A control unit for use in an electronic vapor provision system having a longitudinal axis, the control unit including a control unit engagement end being configured to engage with and disengage from a cartridge including a cartridge engagement end substantially along said longitudinal axis, the control unit comprising:

a battery and a tube portion disposed along said longitudinal axis, wherein the tube portion comprises an inner wall and an outer wall; and

a drive coil being located in the tube portion of the control unit, between the inner wall and the outer wall of the tube portion, and disposed about said longitudinal axis; the drive coil being configured to form an induction heating assembly with a heater element located in the cartridge, wherein the cartridge is located at least partly

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within the inner wall of the tube portion when engaged with said control unit, such that the heater element is within the drive coil,

wherein the control unit is configured such that when the cartridge is engaged with the control unit, a gap is formed between the cartridge engagement end and the control unit engagement end, whereby, in use, air is able to be drawn from the gap through an air inlet of the cartridge provided in the cartridge engagement end; and

wherein the control unit is provided with one or more holes in a housing of the control unit to allow, in use, air to be drawn into the control unit and into the gap.

10. A cartridge for use in an electronic vapor provision system having a longitudinal axis, the cartridge being configured to engage with and disengage from a control unit substantially along said longitudinal axis, the control unit including a control engagement end and the cartridge including a cartridge engagement end, the control unit including a battery and a tube portion disposed along said longitudinal

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axis, wherein the tube portion comprises an inner wall and an outer wall, the cartridge comprising:

a heater element located in the cartridge,

the heater element being configured to form an induction heating assembly with a drive coil located in the tube portion of the control unit, between the inner wall and the outer wall of the tube portion, and disposed about said longitudinal axis, wherein the cartridge is located at least partly within the inner wall of the tube portion when engaged with said control unit, such that the heater element is within the drive coil,

wherein the control unit is configured such that when the cartridge is engaged with the control unit, a gap is formed between the cartridge engagement end and the control unit engagement end, whereby, in use, air is able to be drawn from the gap through an air inlet of the cartridge provided in the cartridge engagement end;

wherein the control unit is provided with one or more holes in a housing of the control unit to allow, in use, air to be drawn into the control unit and into the gap.

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